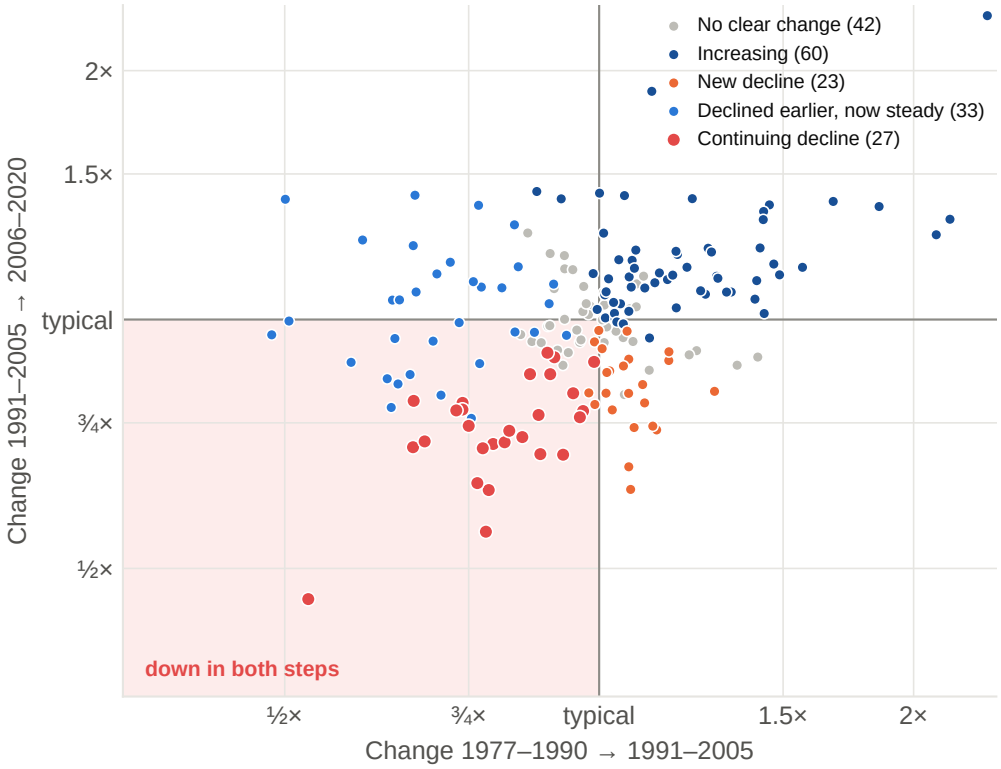
QUESTION 1

Which species, if any, are continuing to decline?

27 native species declined in each period from 1977 to 2020

Change vs typical species: 1977–1990 → 1991–2005 (across) and 1991–2005 → 2006–2020 (up)



Still declining (change 1977→2020)

Hooded Robin	-76%
Pallid Cuckoo	-56%
Little Eagle	-53%
Buff-rumped Thornbill	-51%
Scarlet Robin	-51%
Little Lorikeet	-51%
Dusky Woodswallow	-46%
Restless Flycatcher	-45%
Crested Shrike-tit	-44%
Horsfield's Bronze-Cuckoo	-43%
Jacky Winter	-43%
Rainbow Bee-eater	-42%
Striated Thornbill	-42%
Yellow-billed Spoonbill	-41%
White-naped Honeyeater	-39%
Mistletoebird	-39%
Fuscous Honeyeater	-38%
Brown-headed Honeyeater	-36%
Brown Treecreeper	-32%
Yellow Thornbill	-26%
Dollarbird	-25%
Southern Whiteface	-24%
Gang-gang Cockatoo	-22%
White-throated Treecreeper	-22%
Yellow-rumped Thornbill	-18%
Peaceful Dove	-18%
Yellow-faced Honeyeater	-11%

KEY FINDINGS

- 27 of 185 native species declined in both periods. Steepest: Hooded Robin, Pallid Cuckoo, Little Eagle.
- 18 of the 27 mainly eat insects.
- 23 more show a new decline in 2006–2020; 33 declined earlier and are now steady.

POSSIBLE CAUSES · OUR SPECULATION, NOT TESTED HERE

- Clearing and thinning of woodland, especially in the wheat-sheep belt and tablelands, has left remnants too small and too bare for ground-feeding robins, treecreepers and thornbills. Fallen timber and leaf litter removed by grazing and firewood collection take their insects with them.
- Fewer insects overall, from pesticides, heavy grazing of the ground layer and the Millennium Drought (about 2001–2009), which fell within the last period and would cut food for breeding.
- Noisy Miners spreading through fragmented woodland (see 'Hidden Pressures') push small birds out; cuckoos probably follow the decline of the small birds they rely on as hosts.
- Prediction: without habitat restoration these declines continue after 2020, and the 2019–20 bushfires likely deepen them for forest species such as Gang-gang Cockatoo. Several species on the list are already listed as threatened in NSW, which fits this result.

Atlas surveys 1977–2020; same effort per grid square in each period (432 squares). Change is relative to the typical species. Introduced European Goldfinch and Common Starling also declined and are not listed. Full table: outputs/Q1_species_trends.csv